

Module 9

Materials Deep Dive — Tier 2: Intermediate

Material Comparison

Material	Properties	Best for
PLA	Easy to print, rigid, low strength/heat resistance	General prints, beginners, display pieces
PETG	Tougher, more flexible, better heat resistance than PLA	Functional parts, outdoor use, mechanical parts
ABS	Strong, heat-resistant, prone to warping, needs enclosure	Automotive-style parts, high-heat environments
TPU	Flexible/rubber-like, harder to print (slow speed needed)	Phone cases, gaskets, flexible joints

General Temperature Guidance

- PLA: ~190–220°C nozzle, ~50–60°C bed
- PETG: ~230–250°C nozzle, ~70–80°C bed
- ABS: ~230–250°C nozzle, ~90–110°C bed, enclosure strongly recommended
- TPU: ~220–230°C nozzle, slow print speed required

Always check the specific manufacturer's recommended settings — these vary by brand and colorant.

End of Module 9 — Next: Module 10, Calibration & Tuning